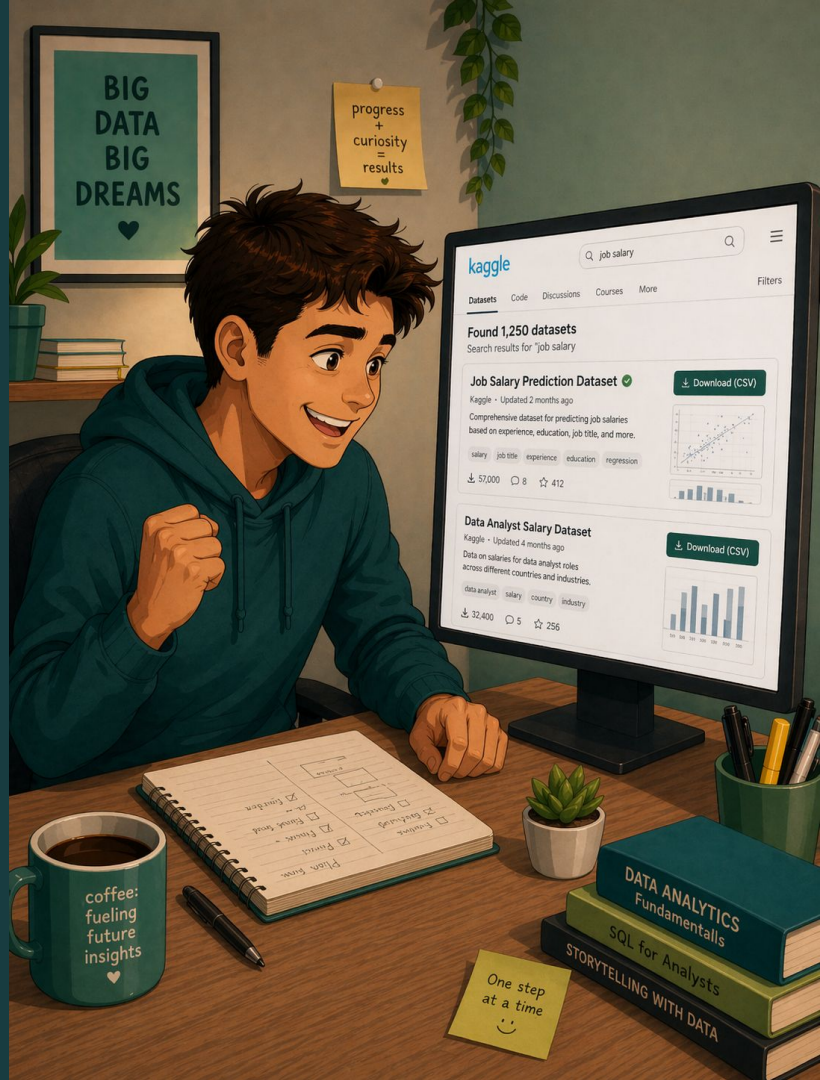
Job Salary Analysis

From Flat CSV to Relational Database and Business Insights

Johannes Vidal

Dataset: Job Salary Prediction Dataset

Source: Kaggle



Dataset and Research Questions

Size: 250,000 records
Structure: Flat CSV with 10 variables

Key Variables:

job_title – professional role

experience_years – years of experience

education_level – academic attainment

remote_work – work arrangement

industry – sector category

salary – yearly USD salary record

Main Hypotheses:

- Experience increases salary
- Education increases salary
- Remote work affects salary
- Industry may influence salary

From Flat CSV to Relational Database

Flat CSV
↓
Split into logical entities
↓
3 related tables
↓
Foreign keys
↓
Exact SQL joins
↓
Analysis-ready database

1. Table creation

```
CREATE TABLE job_profiles(...)
CREATE TABLE work_contexts(...)
CREATE TABLE salary_records (...)
```

2. Define relationships

```
FOREIGN KEY (...) REFERENCES (...)
```

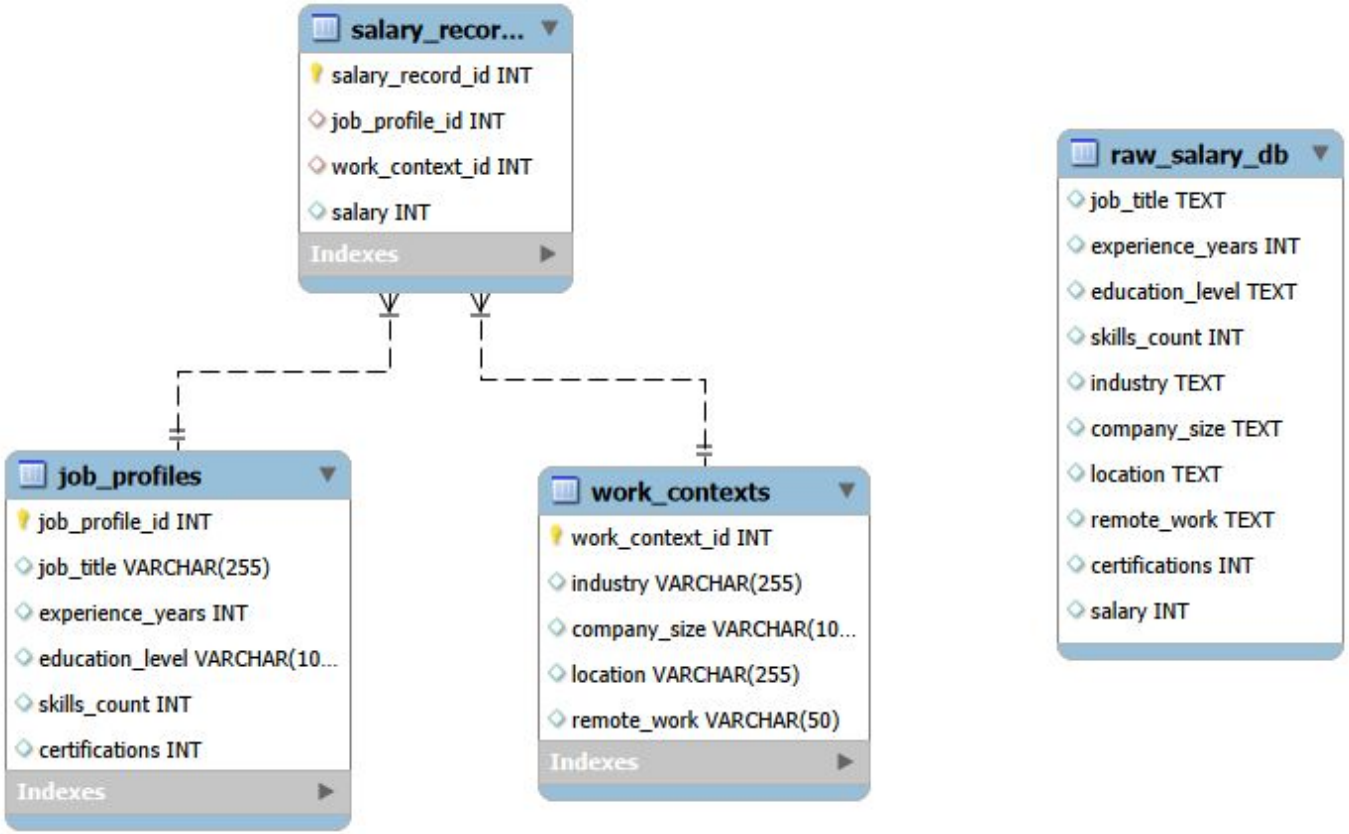
3.Insert unique values

```
INSERT INTO job_profiles (...)
SELECT DISTINCT (...)
```

4. Reconnect with exact joins

```
JOIN job_profiles jp
ON r.job_title = jp.job_title
AND (...)  AND (...)
```

Database Schema



Main Challenge

Turning a flat dataset into logical entities without forcing an unrealistic structure.

This design made the later SQL analysis cleaner and more scalable.

SQL Insights and Advanced Analysis

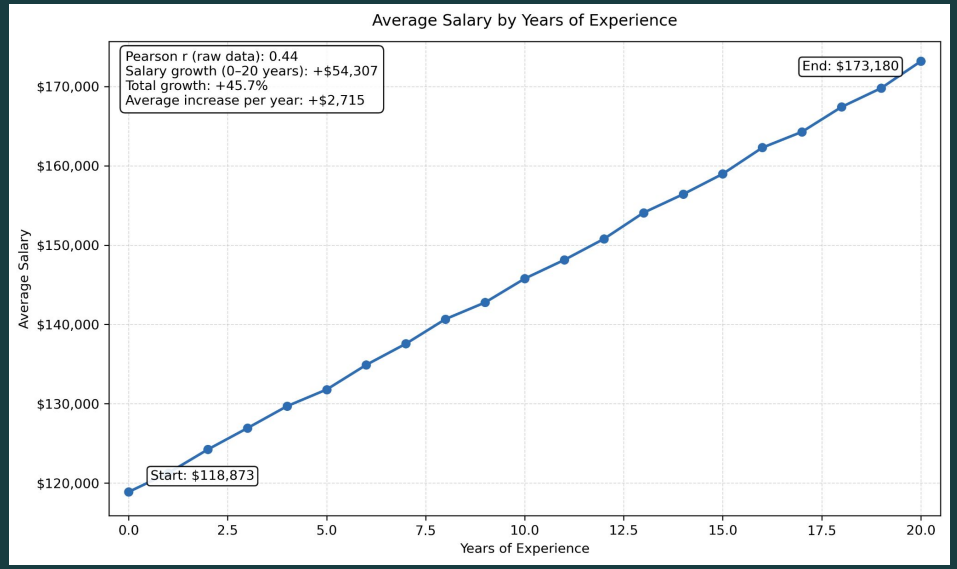
Example query: salary growth by experience

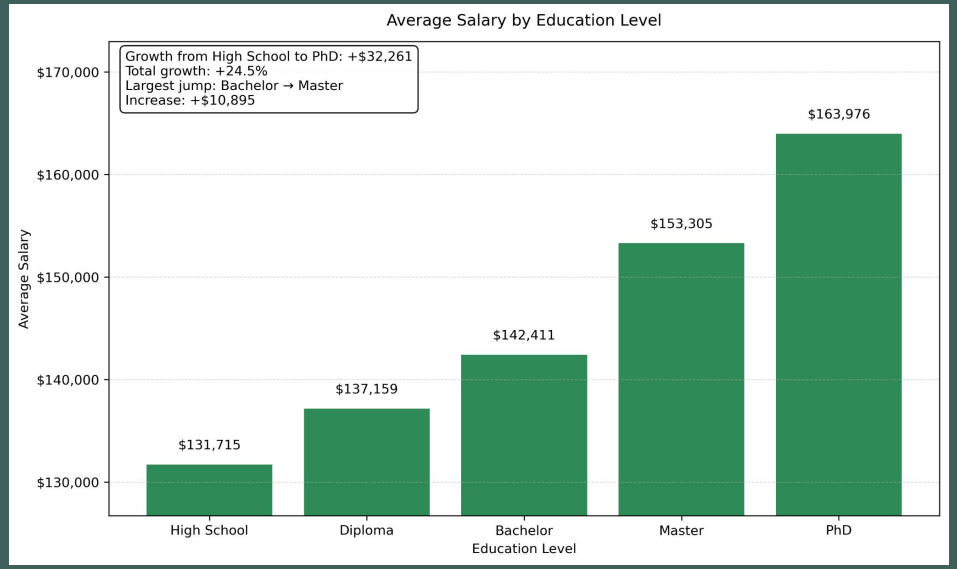
Core SQL logic

- CTE creates a temporary summary table
- JOINs link the relational tables
- GROUP BY calculates average salary by experience
- LAG() shows salary progression over time

```sql
WITH yearly_salary AS
(
    SELECT
        jp.experience_years,
        AVG(sr.salary) AS average_salary
    FROM salary_records sr
    JOIN job_profiles jp
        ON sr.job_profile_id = jp.job_profile_id
    GROUP BY jp.experience_years
)
SELECT
    experience_years,
    average_salary,
    LAG(average_salary) OVER (ORDER BY experience_years)
    AS previous_salary
FROM yearly_salary;
```

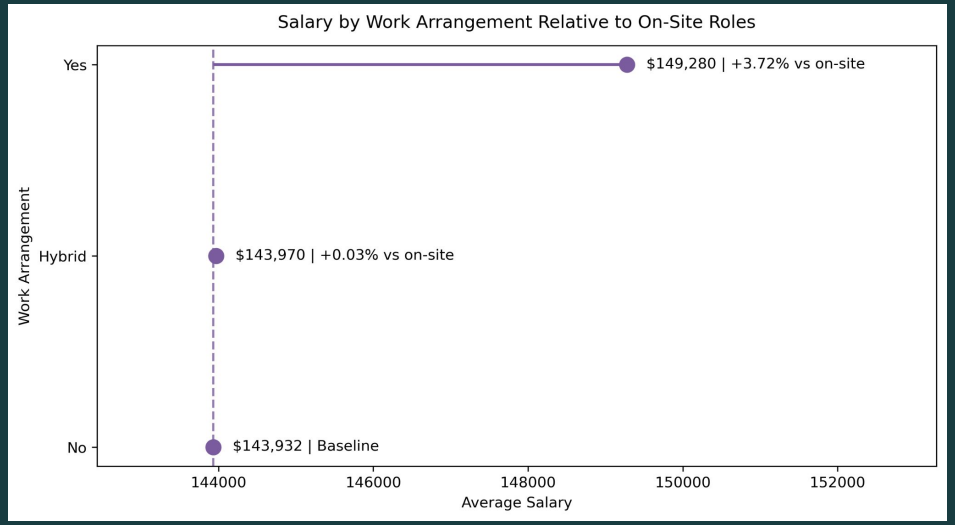
Main Findings: Experience and Education

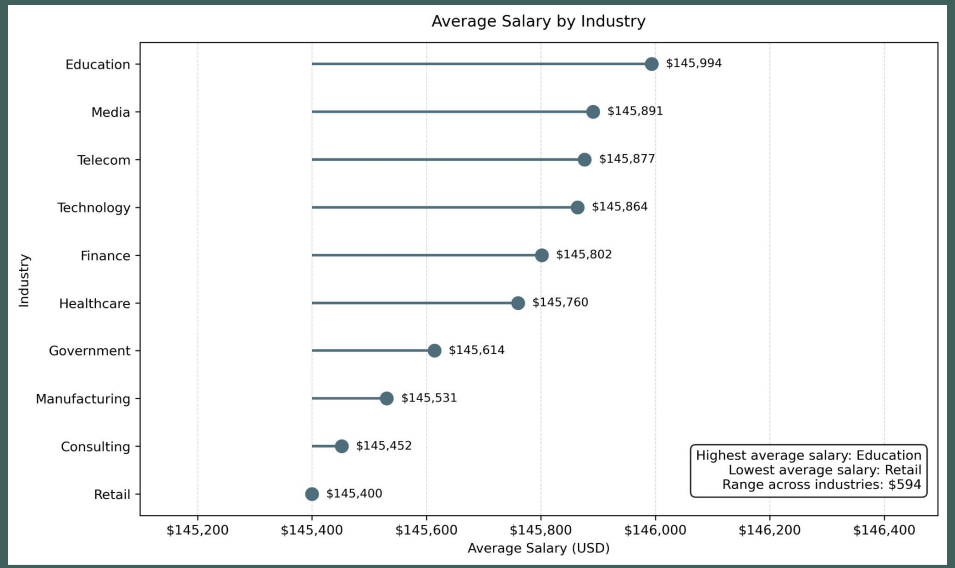




- Experience drives steady salary progression across the career path
- Education creates clearer step changes, especially at postgraduate levels
- Salary growth appears to depend on both accumulated experience and academic qualification

Main Findings: Remote Work and Industry

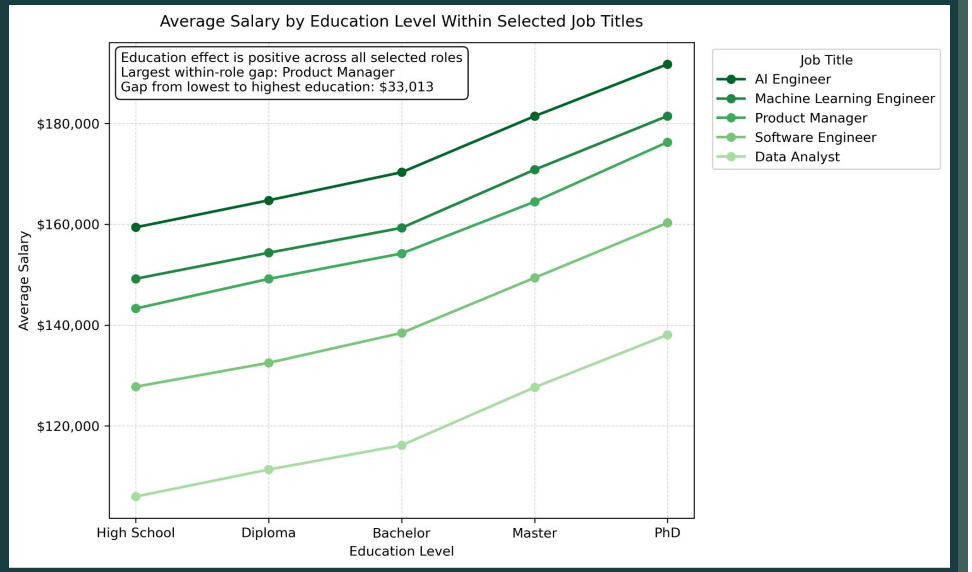


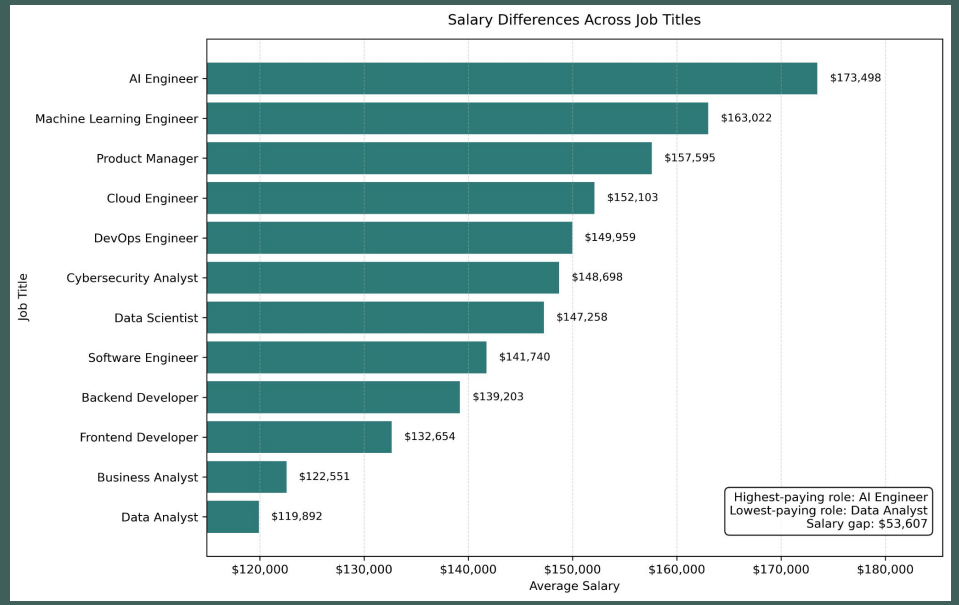


- Fully remote roles show a moderate salary advantage over on-site positions

- Industry shows only minor salary variation across the dataset

Digging Deeper





- The education effect remains visible within every selected role

- Job title emerges as one of the strongest salary differentiators in the dataset

This helps explain why industry showed only weak salary differences

Wrap-Up

- Experience and education were the strongest salary drivers
- Remote work had a moderate effect, while industry had very little
- Deeper analysis showed that job title strongly explains salary differences

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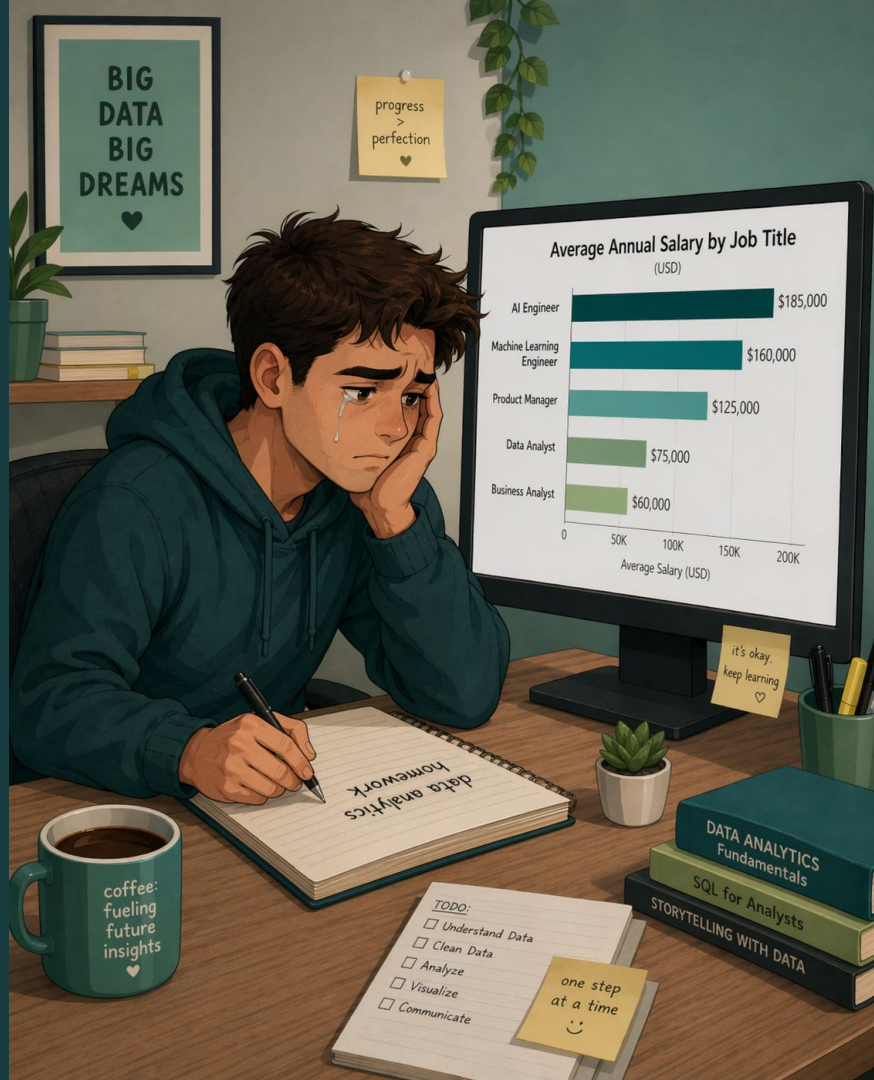
Thank you

Note: Slides showed the core findings and methods only
Full SQL scripts, additional analyses,
and notebook are available in this Github repository